Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

PDA® Research

Survey Results

Transportation Validation Benchmarking

2025

Parenteral Drug Association®

Connecting People, Science and Regulation®

pda.org

©PDA 2025

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Transportation Validation Benchmarking

PDA Survey

2025

ISBN: 978-1-945584-49-7
© 2025 Parenteral Drug Association, Inc.
All rights reserved

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Authors

Co-Leads
Ghada Haddad, PhD
Kite Pharma

Jahanvi Miller
Baxter

Marcia Alves Preda
Nordika Advanced Engineering

Lisa Baker
Amgen

Rafik Bishara, PhD
Eli Lilly and Company (Retired)

Jennifer Eck
Astrazeneca

Jean-Pierre Emond
The Illuminate Group

Gary Hutchinson
Modality Solutions

Sarah Lane
Gilead Sciences Inc.

Amanda Laughlin
Eli Lilly and Company

Felicia Maisuria, MD
St. Georges University

Mark Maurice
Solutions Consulting

Rimple Patwardhan
Eli Lilly and Company

Richard Peck
RP Pharma Consulting

Anthony Rizzo
Cold Chain Technologies

Khoja Suleman Salim
USV Private Limited

Zain Ul Abidin
Drug Regulatory Authority of Pakistan

Kimberly Vazquez
Keros Therapeutics, Inc.

Erik J. van Asselt, PhD
MSD

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Table of Contents

Introduction **1**

General Information (Questions 1.1 – 1.5) **3**

Q 1.1 What type of product(s) does your company manufacture? Select all that apply. 3

Q 1.2 Please indicate whether you are responding to this survey from the perspective of an entire company, a particular manufacturing facility, or multiple manufacturing facilities. 4

Q 1.3 Where are the facilities located you are answering this survey for? Select all that apply. 5

Q 1.4 What is the size of your company? 6

Q 1.5 Please indicate your industry role. 7

Q 1.6 How many qualified shipping lanes does your company operate? 8

Transportation Qualification (Questions 2.1-2.17) **9**

Q 2.1 What method(s) does your organization use to assess the impact of transportation stressors (e.g., vibration, shock/drop, pressure, temperature, and humidity) on the product during transport qualification? Please select all that apply. 9

Q 2.2 How does your company ensure representation of multiple seasons in real-world shipping qualifications for assessing product quality under various transportation stresses? Please select all that apply. 10

Q 2.3 Has your company's simulated transportation qualification approach ever been challenged by a regulatory agency (simulated data only)? 11

Q 2.4 Have any market applications from your company been rejected as a result of this approach (simulated data only)? 12

Q 2.5 Has your company received any post-approval requests to conduct a real-world study for transportation qualification? 13

Q 2.6 Has your company encountered similar regulatory challenges related to transport validation studies or data in any of the following locations? Please select all that apply. 14

Q 2.7 What regulatory agencies does your organization submit data to? Please select all that apply. 15

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.8 What standards does your company adhere to for transportation qualification studies? Please select all that apply..... 16

Q 2.9 Does your company have mileage limits and/or duration limits based on simulated studies?..... 18

Q 2.10 For shipping solutions qualification, does your company perform the following? 19

Q 2.11 How does your company justify the frequency of its monitoring activities? Please select all that apply. 20

Q 2.12 What systems does your company use to gather data? Please select all that apply..... 21

Q 2.13 What type of shipping systems have you been using for real-world qualifications? Please select all that apply..... 22

Q 2.14 Has a regulatory agency challenged your company's placement of temperature recording devices within the shipping container for routine shipments?..... 23

Q 2.15 Has your organization received requests for real-world temperature data for shipments post-approval, prior to sending them to a specific region or country? 24

Q 2.16 As part of route qualification, has your organization prepared and conducted qualifications for passive or active transport equipment?..... 25

Q 2.17 Would it be beneficial to maintain our efforts in industry engagement, benchmarking, and collaboration with agencies concerning transport qualification studies? 26

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Introduction

This survey aimed to gather industry benchmarking data on transportation qualification strategies within the pharmaceutical and biopharmaceutical industry. Specifically, the goal was to examine the application of simulated data to better understand how transportation conditions impact drug product quality, emphasizing the importance of ensuring product quality during transit. The survey focused on evaluating transportation through simulated environments versus real-world qualification of temperature-controlled shipping systems.

The overarching goal is to evaluate the use of simulated data as a reliable source for regulatory submissions as opposed to conducting real-world studies, which require time and resources that can often delay approvals. Ultimately, those delays can prevent patients from receiving critical medicines in a timely manner. In addition, through the results of this survey, the industry aims to identify opportunities for improving supply-chain efficiency through the use of predictive models. With the increasing use of digitization and artificial intelligence across multiple industries, more consideration should be given to the use of predictive models.

To contextualize responses, the survey included general demographic questions and was structured into product-specific, supply-chain-focused, and regulatory strategy-driven categories. The manufacturers of small molecule medicines and oral dosage forms were not a target audience for this survey.

Overview

This survey was constructed with a set of general demographic questions to better understand the context of the responses from respondents. The survey was also broken down into categories which were product specific, supply chain focused and driven by regulatory strategy.

Summary of Results

Of the 154 respondents to this survey, most were from the pharmaceutical (about 37%) and biopharmaceutical (about 35%) medicines industry, with some participation from the medical device sector. Responses were fairly evenly split between company-wide and site-specific facilities. Geographically, the majority of respondents were manufacturers from the United States and the European Union, with notable participation from India and China as well. The responses were obtained primarily from medium-to-large companies, with a significant representation from large organizations, operating over 100 shipping lanes.

Methods for assessing the impact of transportation stressors (e.g., vibration, shock/drop, pressure, temperature, humidity) during transport qualification were almost equally distributed among the categories presented: Actual Shipping Route (real-world transportation), Route Temperature Mapping,

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Simulated Laboratory Studies, and Worst-Case Shipping Route simulations. Many companies ensure seasonal representation in real-world shipping qualifications by employing controlled environmental conditions to replicate seasonal temperature and humidity variations, temperature mapping of challenging distribution routes, and season-specific testing protocols.

Notably, almost half of the companies utilizing simulated transportation-qualification methods reported facing challenges from regulatory agencies and/or receiving post-approval requests for real-world studies. In contrast, respondents indicated fewer regulatory challenges with real-world transport data methodologies, particularly for those qualifying passive or active transport equipment.

Conclusions

The significant number of industry respondents highlights a clear need for the consideration of simulated transport data in regulatory submissions and the desire for uniformly drafted future guidance on transportation-qualification practices. The fact that almost 50% of companies using simulated approaches have faced regulatory challenges underscores the necessity for clarification and standardization in this area. Post-approval requests for real-world studies from the U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA), Pharmaceuticals and Medical Devices Agency (PDMA) and other regulators can cause significant disruptions due to resource demands, potentially delaying the supply chain. To mitigate these issues, companies must ensure compliance, develop risk-based strategies, enhance oversight, and leverage real-time data collection and analysis for a more dynamic approach.

Concerns were also expressed regarding anticipated regulatory responses to simulated data submissions. Understanding past agency responses could facilitate the development of comprehensive methodologies and qualification programs for simulated-transport submissions across the industry. Overwhelmingly, respondents agreed on the benefit of continued industry engagement, benchmarking, and collaboration with regulatory agencies concerning transportation-qualification studies.

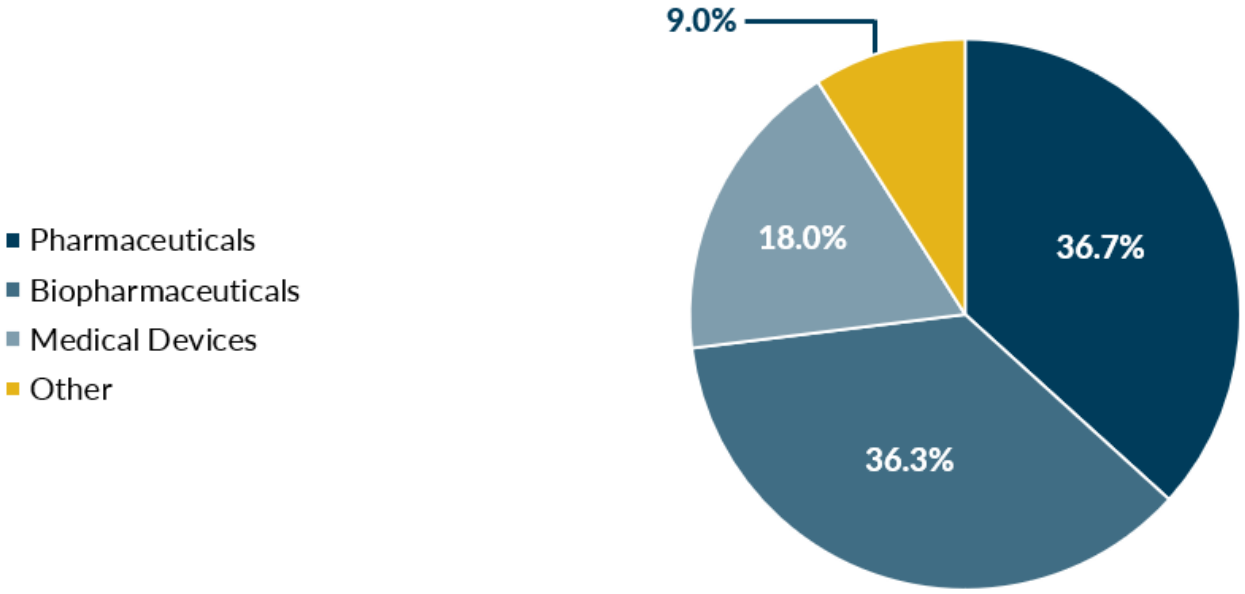
 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

General Information (Questions 1.1 – 1.5)

Q 1.1 What type of product(s) does your company manufacture? Select all that apply.

Total Respondents: 154 *Total Responses: 256*



Product Type	Responses	Percent
Pharmaceuticals	94	36.7%
Biopharmaceuticals	93	36.3%
Medical Devices	46	18.0%
Other*	23	9.0%
Total	**256**	**100.0%**

* Twenty-three responses were shared under the category "Other":

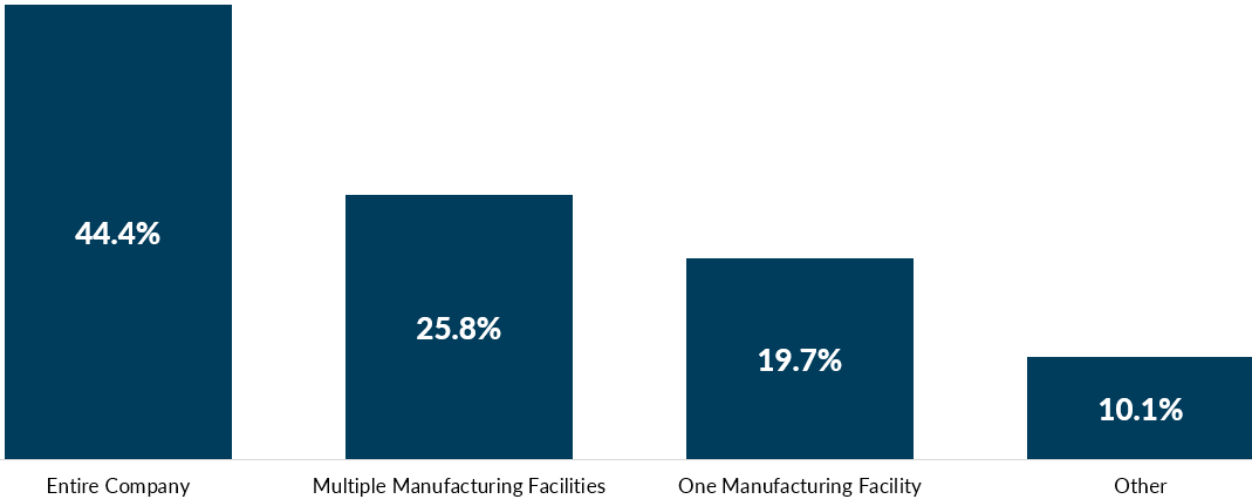
- Advanced Therapy Medicinal Products (ATMPs) – **4 Responses**
- Thermal Packaging **– 4 Responses**
- Combination Products – **3 Responses**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

- Consultant – **3 Responses**
- Testing Laboratory – **2 Responses**
- Transportation Logistics – **2 Responses**
- Injectables
- None
- Primary Packaging
- Temperature and Humidity Monitoring Solutions
- Viral Vectors

Q 1.2 Please indicate whether you are responding to this survey from the perspective of an entire company, a particular manufacturing facility, or multiple manufacturing facilities.

Total Respondents: 154 *Total Responses: 154*



Response Selection	Responses	Percent
Entire Company	68	44.4%
Multiple Manufacturing Facilities	42	25.8%
One Manufacturing Facility	30	19.7%
Other*	14	10.1%
Total	**154**	**100.0%**

* Fourteen responses were shared under the category "Other":

- As Quality Manager
- As Regulator
- Contract Lab, Single Facility

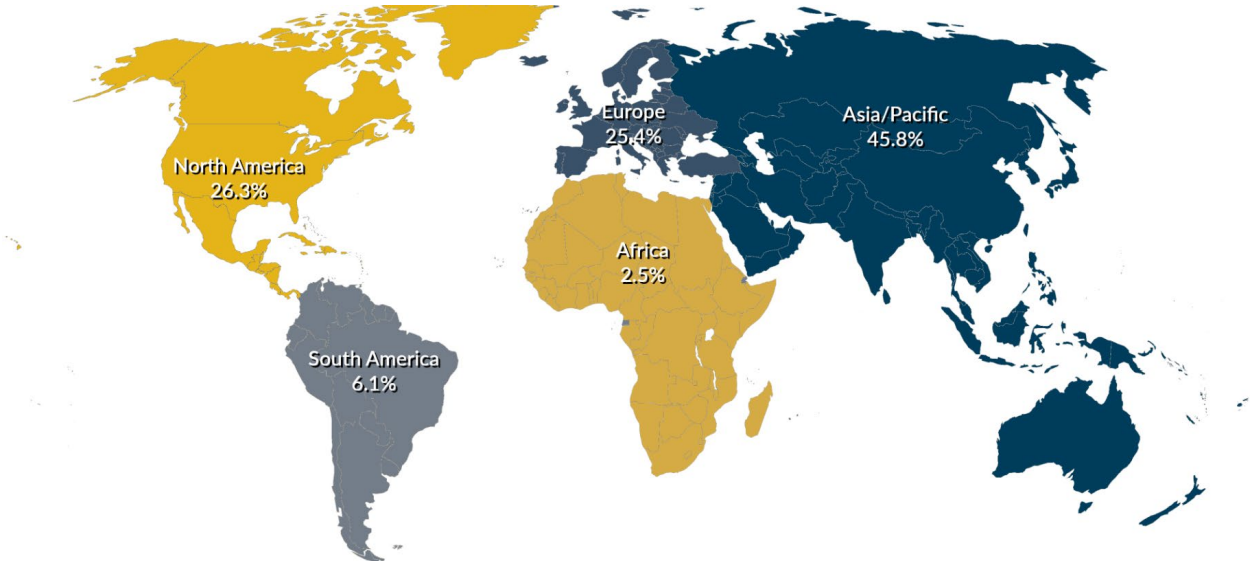
 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

- Development Unit
- Drug Regulator
- From My Knowledge of Related Clients
- GDP Transportation Logistics Company
- Global Group Covering External Manufacturing for One Dosage Form
- Global Technical Operations
- I am responding as myself who has over 20 years working in cold chain for large BioTech and C&GT companies, and who also was a founder of simulating transport on product.
- Individual
- My Client Experience
- My Own Perspective
- One Development Business Unit
- Own Experience
- QA
- R&D
- Testing Lab

Q 1.3 Where are the facilities located you are answering this survey for? Select all that apply.

Total Respondents: 154 *Total Responses: 441*

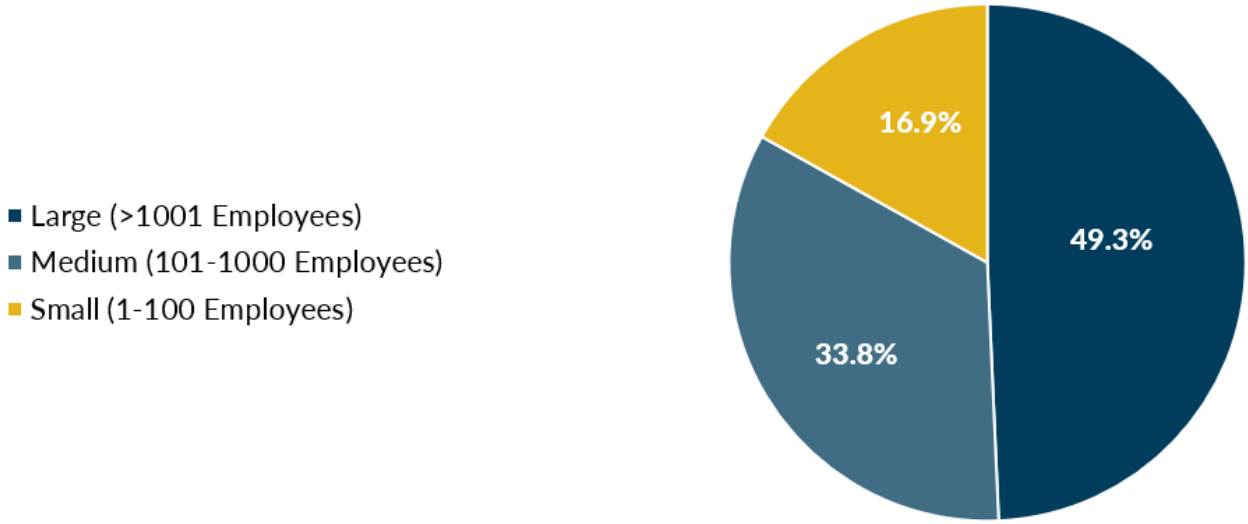


Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Country	Responses	Percent
North America	116	26.3%
Europe	112	25.4%
India	39	8.8%
China	34	7.7%
Japan	32	7.3%
Latin/South America	27	6.1%
Australia/New Zealand	21	4.8%
Middle East	21	4.8%
Other Asian Country Not Listed	20	4.5%
Africa	11	2.5%
Russia	8	1.8%
Total	**441**	**100.0%**

Q 1.4 What is the size of your company?

Total Respondents: 154 *Total Responses: 154*



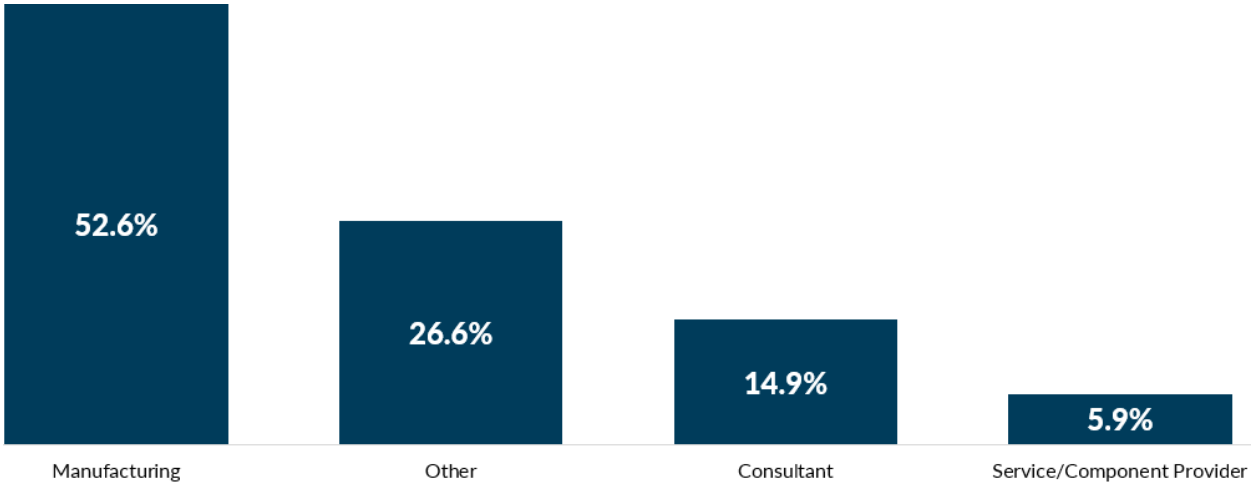
 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Company Size	Responses	Percent
Large (>1001 Employees)	76	49.3%
Medium (101-1000 Employees)	52	33.8%
Small (1-100 Employees)	26	16.9%
Total	**154**	**100.0%**

Q 1.5 Please indicate your industry role.

Total Respondents: 154 *Total Responses: 154*



Industry Role	Responses	Percent
Manufacturing	81	52.6%
Other*	41	26.6%
Consultant	23	14.9%
Service/Component Provider	9	5.9%
Total	**154**	**100.0%**

* Forty-one responses were shared under the category "Other":

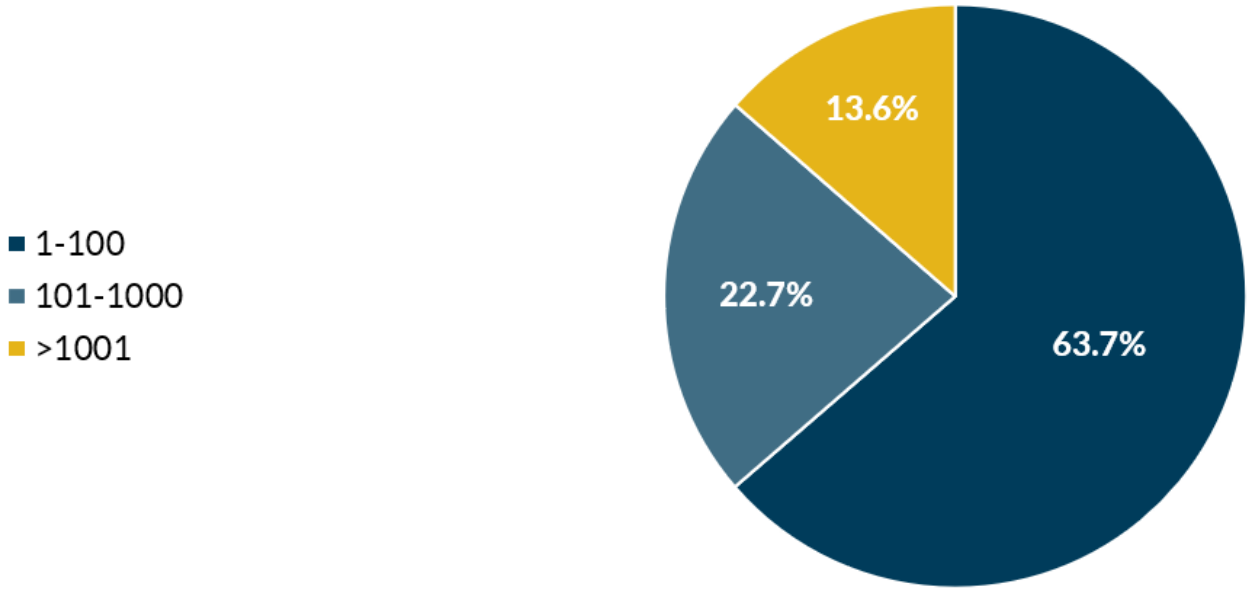
- Quality – **17 Responses**
- Technical Operation – **6 Responses**
- Logistics – **4 Responses**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

- Regulators – **4 Responses**
- Research and Development (R&D) – **4 Responses**
- Engineer/Engineering – **2 Responses**
- Package Development (Sr. Leadership)
- Chief Executive Officer (CEO)
- Product Management
- Manufacturer, Consultant and Service Provider

Q 1.6 How many qualified shipping lanes does your company operate?

Total Respondents: 154 *Total Responses: 154*



Qualified Shipping Lanes	Responses	Percent
1-100	98	63.7%
101-1000	35	22.7%
>1001	21	13.6%
Total	**154**	**100.0%**

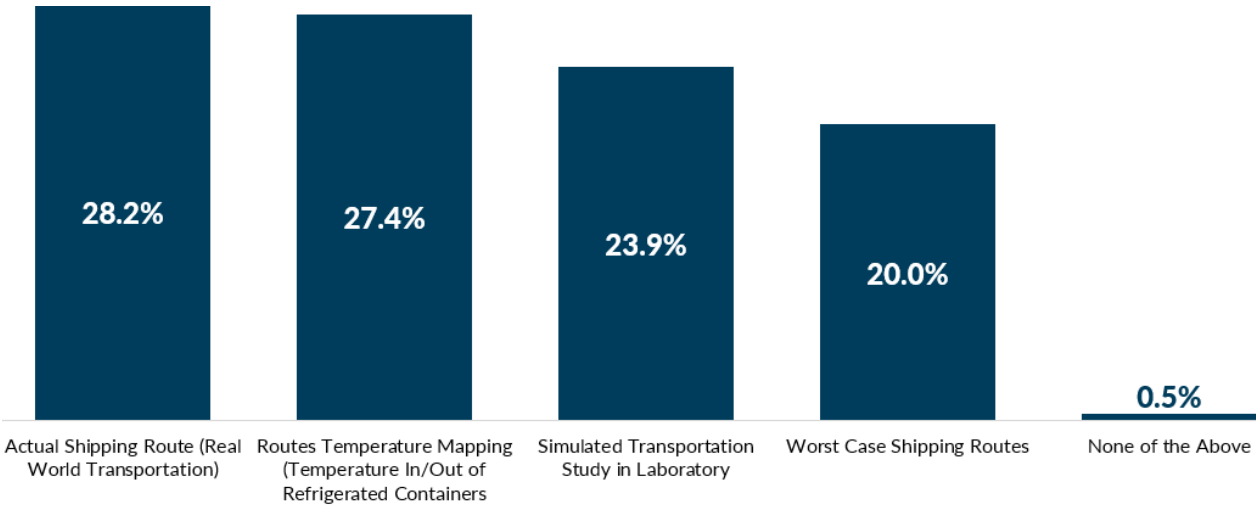
 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Transportation Qualification (Questions 2.1-2.17)

Q 2.1 What method(s) does your organization use to assess the impact of transportation stressors (e.g., vibration, shock/drop, pressure, temperature, and humidity) on the product during transport qualification? Please select all that apply.

Total Respondents: 154 *Total Responses: 394*

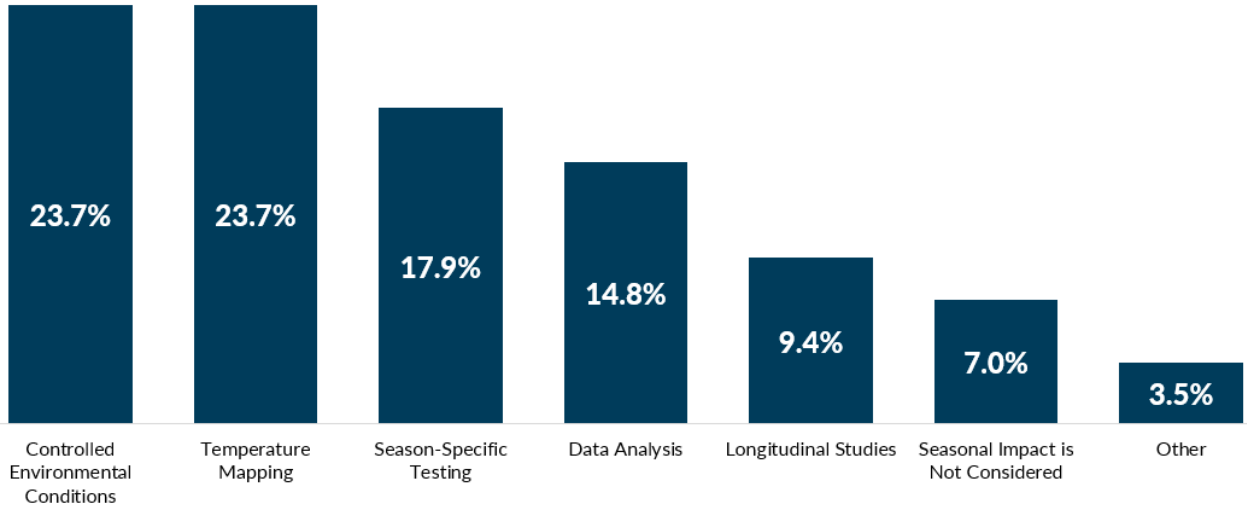


Transportation Stressor Assessment Method	Responses	Percent
Actual Shipping Route (Real World Transportation)	111	28.2%
Routes Temperature Mapping (Temperature In/Out of Refrigerated Containers	108	27.4%
Simulated Transportation Study in Laboratory	94	23.9%
Worst Case Shipping Routes	79	20.0%
None of the Above	2	0.5%
Total	**394**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.2 How does your company ensure representation of multiple seasons in real-world shipping qualifications for assessing product quality under various transportation stresses? Please select all that apply.

Total Respondents: 107 *Total Responses: 257*



Multiple Season Representation Approach	Responses	Percent
Controlled Environmental Conditions: Using controlled environments in testing facilities to replicate seasonal conditions (e.g., temperature and humidity variations) during transport qualification.	61	23.7%
Temperature Mapping: Relies on temperature mapping conducted for the most challenging routes within your company's distribution network.	61	23.7%
Season-Specific Testing: Implementing separate testing protocols for different seasons, simulating conditions such as winter cold or summer heat.	46	17.9%
Data Analysis: Analyzing historical shipping data to understand how products have performed in different seasonal conditions and using that information to inform testing protocols.	38	14.8%
Longitudinal Studies: Conducting shipping trials over extended periods that span different seasons to gather data across varying conditions.	24	9.4%
Seasonal Impact is Not Considered	18	7.0%
Other*	9	3.5%
Total	**257**	**100.0%**

 © 2025 Parenteral Drug Association, Inc.

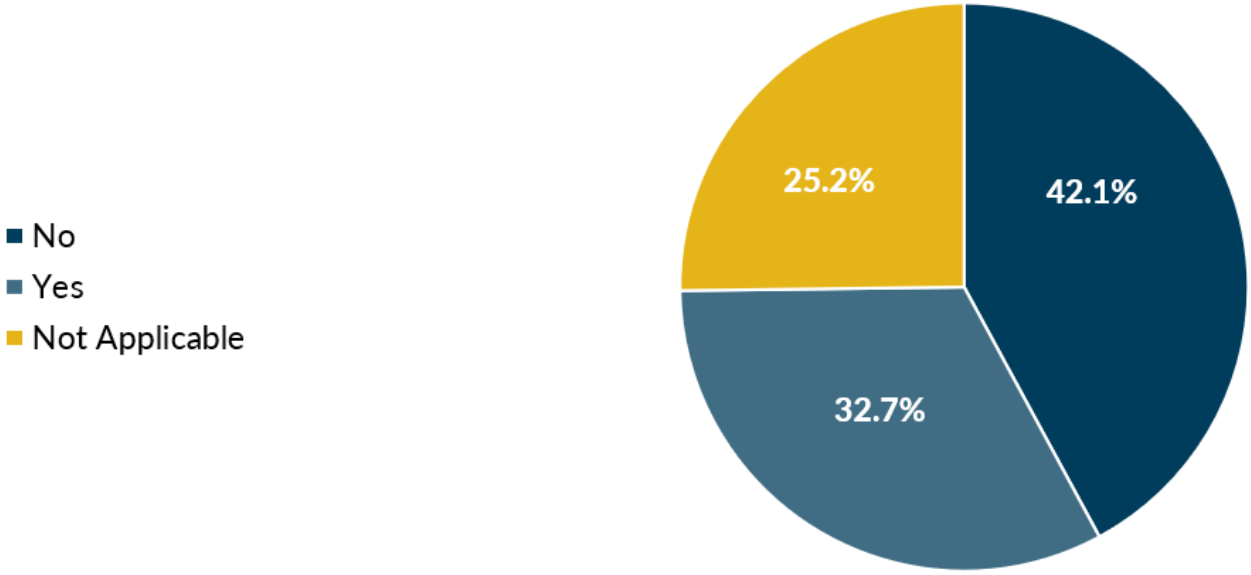
Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

* Nine responses were shared under the category "Other":

- 100% temperature monitoring on routine shipments
- All shipments temperature monitoring +/- 1 degree C
- Definition of climate classifications
- Different approaches are used according to location (mfg site, operating company)
- If the question refers to thermal shipping qualification, we include simulated data (OQ) that captures seasonal impact and PQ provides real world support of the qualified thermal shipping systems.
- It depends on the criticality of the product and the need to save money vs. cannot fail. C&GT cannot fail, so we require robust shipping systems since you only get 1 shot. In the end, its understanding your specific company's needs, documenting your approach, and why its your approach. What works for 1 company or product may not work for another.
- Products are shipped under controlled conditions and documented.
- The Brazilian Market Winter and Summer studies performed using Temperature mapping only
- Utilize qualification work done by vendors of active containers to inform testing protocols.

Q 2.3 Has your company's simulated transportation qualification approach ever been challenged by a regulatory agency (simulated data only)?

Total Respondents: 107 *Total Responses: 107*

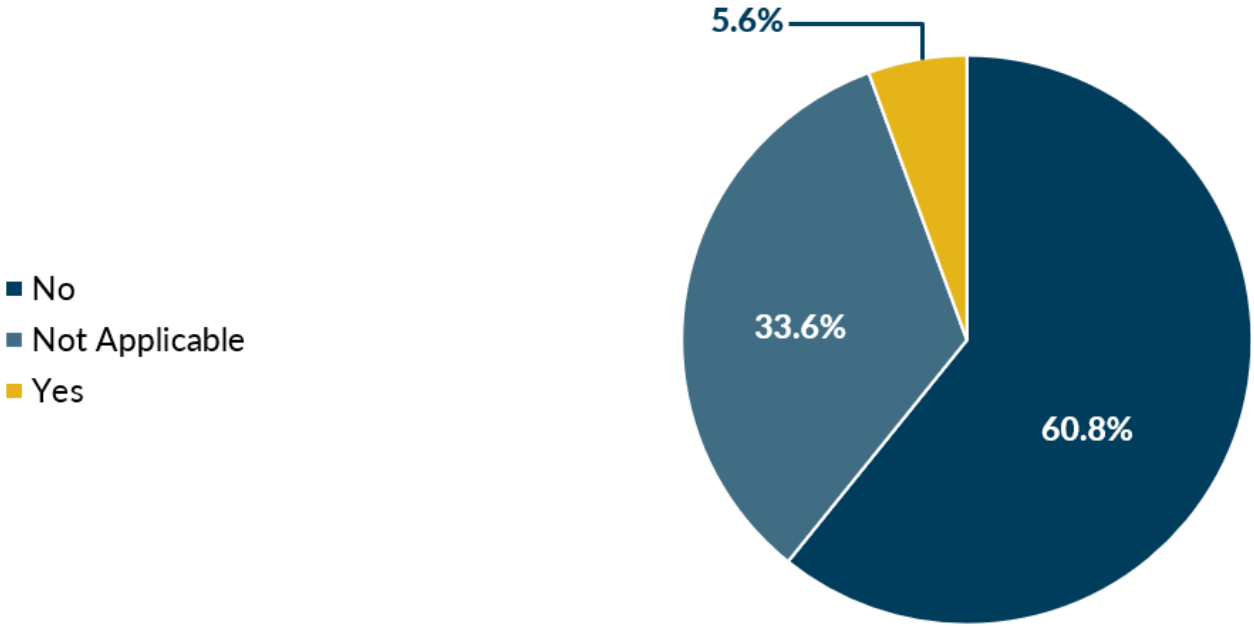


Approach Challenged	Responses	Percent
No	45	42.1%
Yes	35	32.7%
Not Applicable	27	25.2%
Total	**107**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.4 Have any market applications from your company been rejected as a result of this approach (simulated data only)?

Total Respondents: 107 *Total Responses: 107*



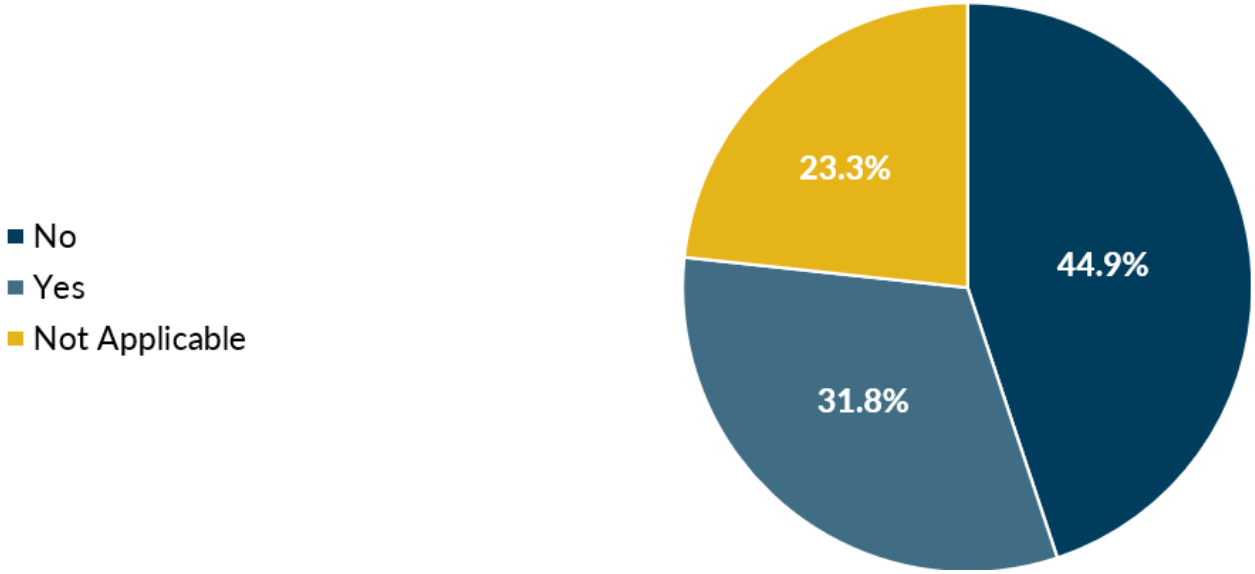
Applications Rejected	Responses	Percent
No	65	60.8%
Not Applicable	36	33.6%
Yes	6	5.6%
Total	**107**	**100.0%**

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.5 Has your company received any post-approval requests to conduct a real-world study for transportation qualification?

Total Respondents: 107 *Total Responses: 107*

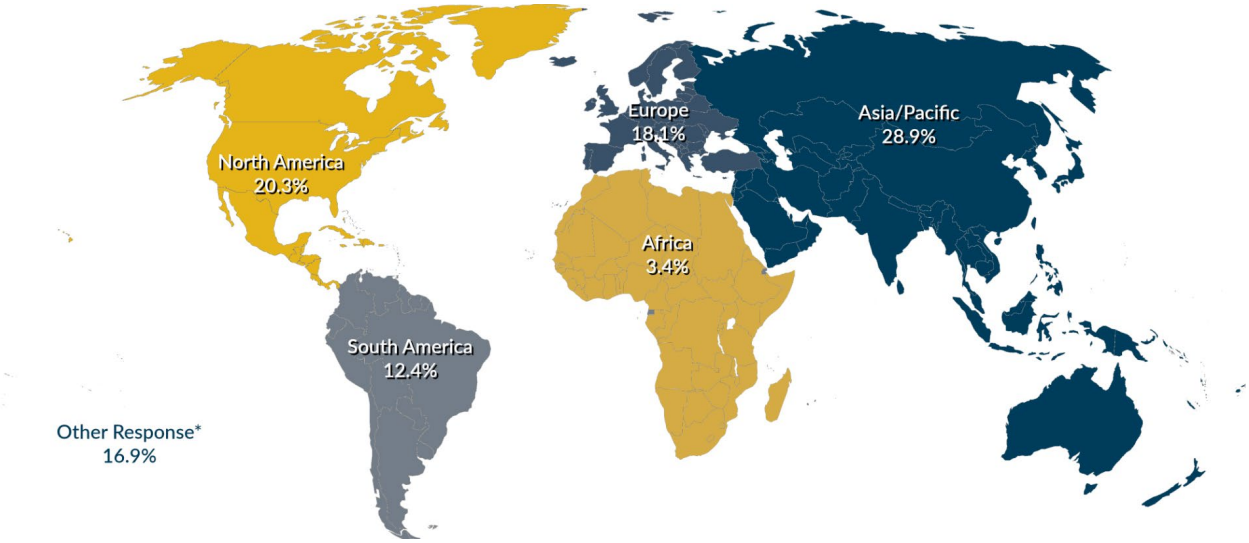


Post-Approval Requests	Responses	Percent
No	48	44.9%
Yes	34	31.8%
Not Applicable	25	23.3%
Total	**107**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.6 Has your company encountered similar regulatory challenges related to transport validation studies or data in any of the following locations? Please select all that apply.

Total Respondents: 107 *Total Responses: 177*



Country	Responses	Percent
North America	36	20.3%
Europe	32	18.1%
Other Response*	30	16.9%
Latin/South America	22	12.4%
Japan	11	6.2%
Other Asian Country Not Listed	11	6.2%
Australia/New Zealand	9	5.1%
China	7	4.0%
India	7	4.0%
Africa	6	3.4%
Middle East	4	2.3%

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

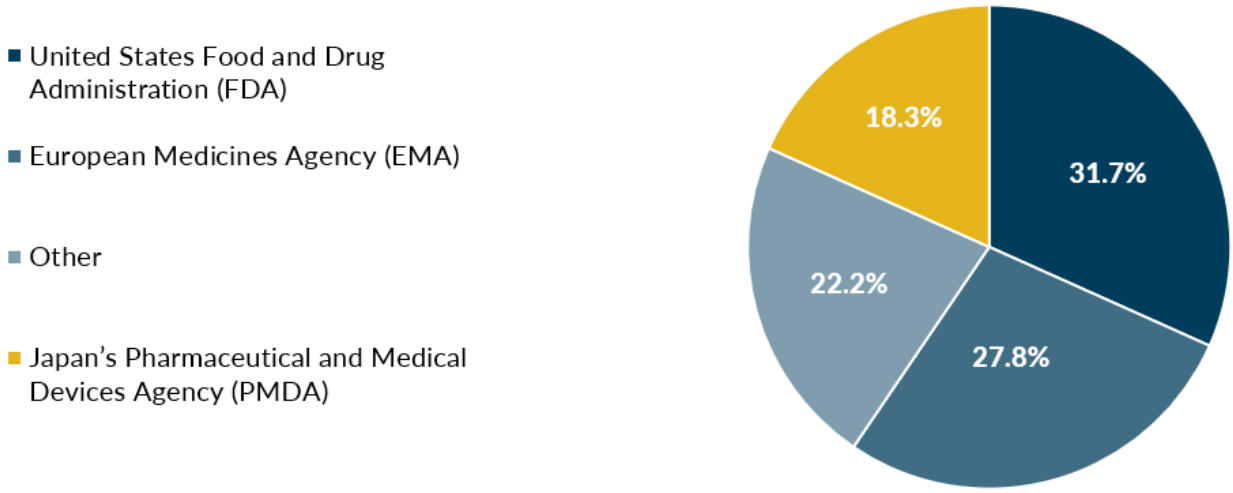
Russia	2	1.1%
Total	**177**	**100.0%**

* Thirty responses were shared under the category "Other Response":

- No; None – **18 Responses**
- Not Applicable (N/A); Does Not Apply – **7 Responses**
- Not Aware; Not Sure – **4 Responses**
- This question is not clear to me. Similar to what? The subjects questions 9-11? Similar from region to region?

Q 2.7 What regulatory agencies does your organization submit data to? Please select all that apply.

Total Respondents: 107 *Total Responses: 252*



Regulatory Agency	Responses	Percent
United States Food and Drug Administration (FDA)	80	31.7%
European Medicines Agency (EMA)	70	27.8%
Other*	56	22.2%
Japan's Pharmaceutical and Medical Devices Agency (PMDA)	46	18.3%
Total	**252**	**100.0%**

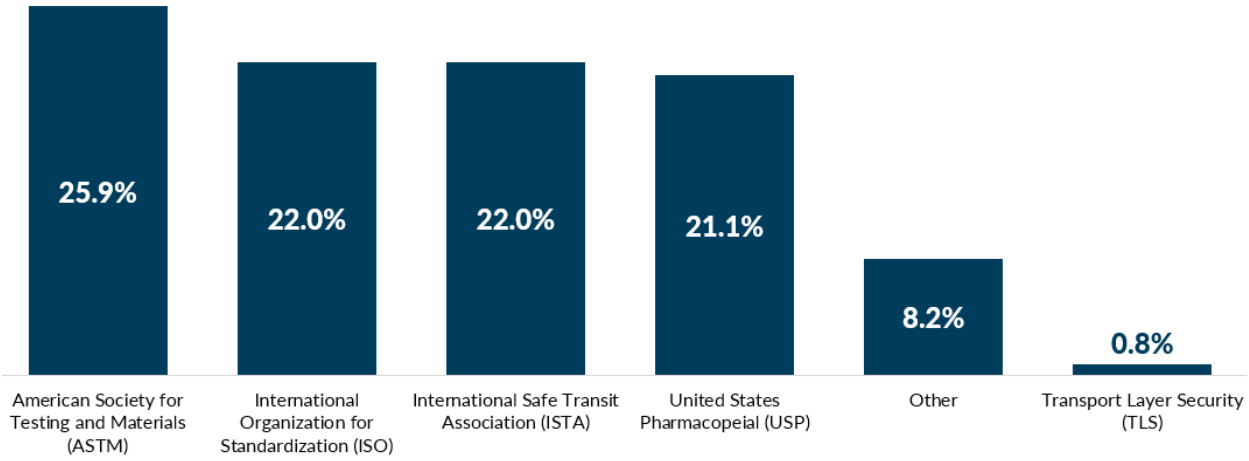
Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

* Fifty-six responses were shared under the category "Other":

- Brazilian Health Regulatory Agency (ANVISA) – **8 Responses**
- None; Not Applicable (N/A) – **4 Responses**
- China's National Medicinal Products Administration (NMPA) – **3 Responses**
- Global/Worldwide – **3 Responses**
- Health Canada – **3 Responses**
- The Therapeutic Goods Administration of Australia (TGA) – **3 Responses**
- United Kingdom's Medicines and Healthcare Products Regulatory Agency (MHRA) – **3 Responses**
- All Regulatory Agencies Listed – **2 Responses**
- Central Gulf Committee for Drug Regulation (GCC) – **2 Responses**
- Colombia's National Food and Drug Surveillance Institute (INVIMA) – **2 Responses**
- Drug Regulatory Authority of Pakistan (DRAP) – **2 Responses**
- Health Products Regulatory Authority of Ireland (HPRA) – **2 Responses**
- Saudi (KSA) Food and Drug Authority (SFDA) – **2 Responses**
- African Medicines Agency (AMA)
- Additional countries we market to
- Country Specific According to the Market
- Depends
- General Directorate of Medicines, Supplies, and Drugs (DIGEMID) in Peru
- Ghana Food and Drug Authority (GFDA)
- Indonesian Food and Drug Control Agency (BPOM)
- It is summarized in MA and submitted as asked.
- Indian Central Drugs Standard Control Organization (CDSCO)
- Isreal's Ministry of Health (MOH)
- Maritime Area Regulatory Authority (MARA) in Ireland
- Mexican Federal Committee for Protection from Sanitary Risks (COFEPRIS)
- PIC/S countries
- Public Health Institute of Chile (ISP)
- Singapore's Health Sciences Authority (HSA) and others
- Veterinary Medicines Directorate (VMD) in the United Kingdom
- World Health Organization (WHO)

Q 2.8 What standards does your company adhere to for transportation qualification studies? Please select all that apply.

Respondents: 107 *Responses: 232*



 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Standard	Responses	Percent
American Society for Testing and Materials (ASTM)	60	25.9%
International Organization for Standardization (ISO)	51	22.0%
International Safe Transit Association (ISTA)	51	22.0%
United States Pharmacopeial (USP)	49	21.1%
Other*	19	8.2%
Transport Layer Security (TLS)	2	0.8%
Total	**232**	**100.0%**

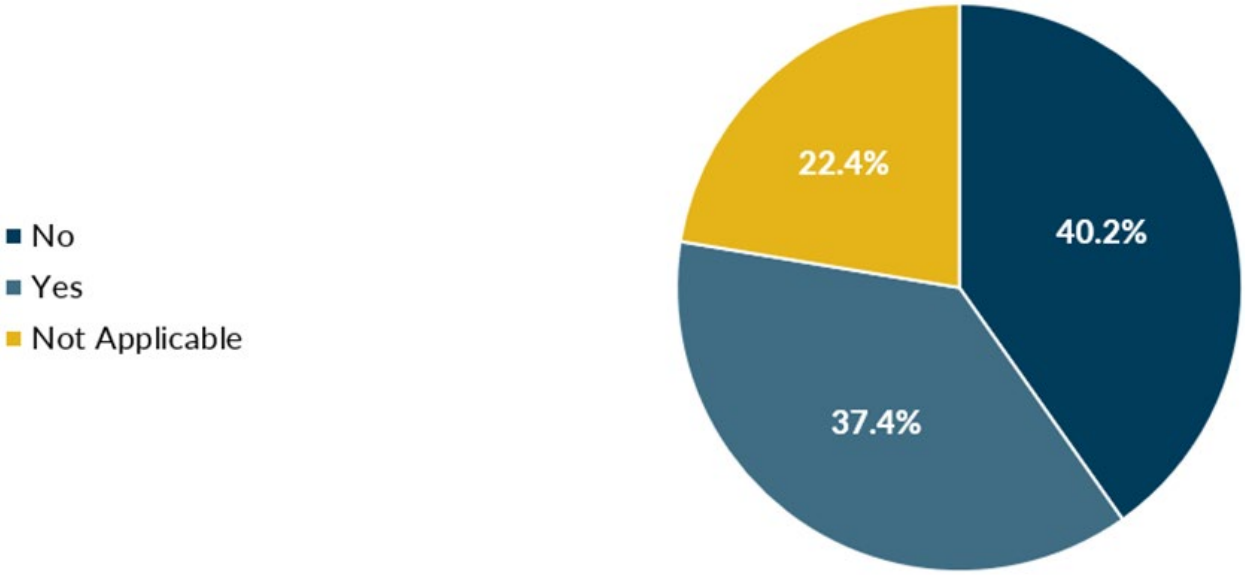
* Nineteen responses were shared under the category "Other":

- World Health Organization (WHO) – **5 Responses**
- A combination of custom and ISTA. Even ISTA states you should understand your environment, so in my view its critical to dive deep into a companies supply chain network and understand it. Where most critical, develop your own standard, where applicable or when "good enough" use a trusted standard, but with data to back up the decision.
- Annex 15/ PDA39 & 64. BPOG Best practice industry doc
- Corporate policies
- Custom
- EU GDP, PDA
- FAA and EASA
- For thermal shipping qualification, we follow internal standards based on interpretation of health authorities expectations
- IATA CEIV Pharma
- ISPE
- ISPE Cold chain Validation guide
- Not all in one standard. parts of several are used, including PDA TRs.
- Not applicable
- PDA, WHO, Anvisa
- Product specific tests

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.9 Does your company have mileage limits and/or duration limits based on simulated studies?

Respondents: 107 *Responses: 107*



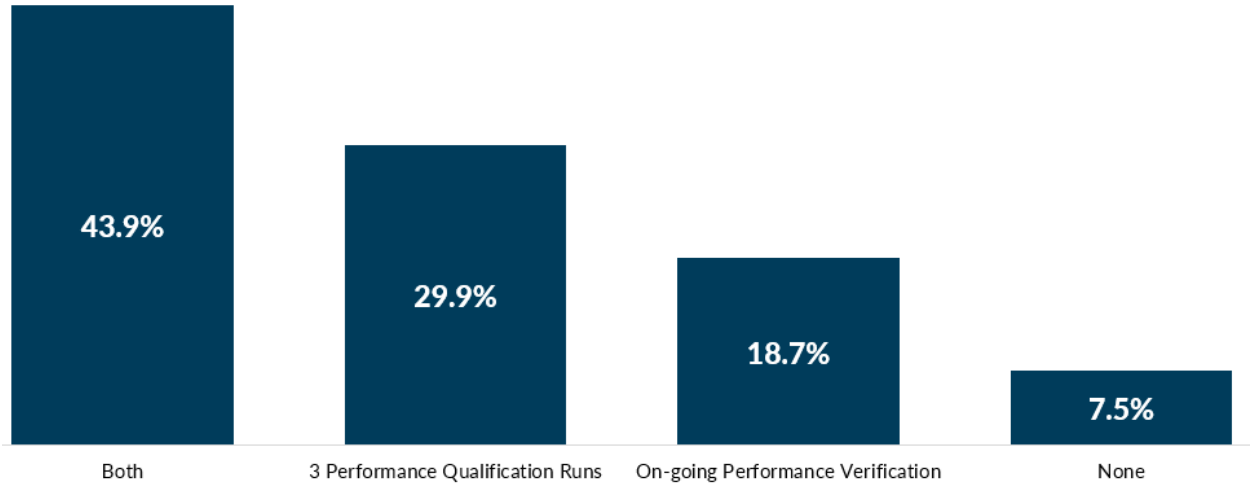
Mileage Limits and/or Duration Limits	Responses	Percent
No	43	40.2%
Yes	40	37.4%
Not Applicable	24	22.4%
Total	**107**	**100.0%**

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.10 For shipping solutions qualification, does your company perform the following?

Total Respondents: 107 *Total Responses: 107*

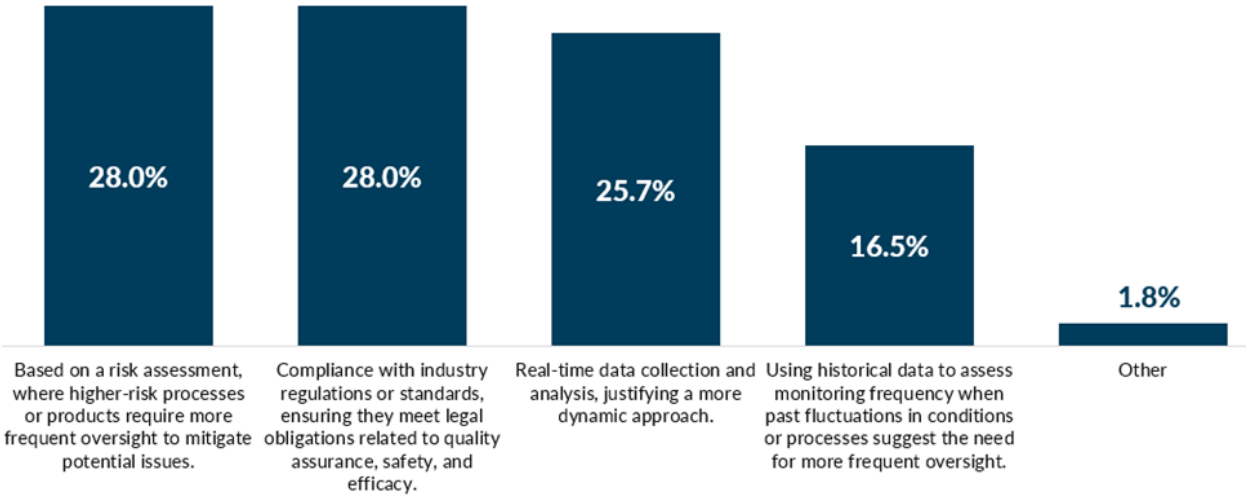


Shipping Solutions Qualification	Responses	Percent
Both	47	43.9%
3 Performance Qualification Runs	32	29.9%
On-going Performance Verification	20	18.7%
None	8	7.5%
Total	**107**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.11 How does your company justify the frequency of its monitoring activities? Please select all that apply.

Total Respondents: 107 *Total Responses: 218*



Frequency Justification	Responses	Percent
Based on a risk assessment, where higher-risk processes or products require more frequent oversight to mitigate potential issues.	61	28.0%
Compliance with industry regulations or standards, ensuring they meet legal obligations related to quality assurance, safety, and efficacy.	61	28.0%
Real-time data collection and analysis, justifying a more dynamic approach.	56	25.7%
Using historical data to assess monitoring frequency when past fluctuations in conditions or processes suggest the need for more frequent oversight.	36	16.5%
Other*	4	1.8%
Total	**218**	**100.0%**

* Four responses were shared under the category "Other":

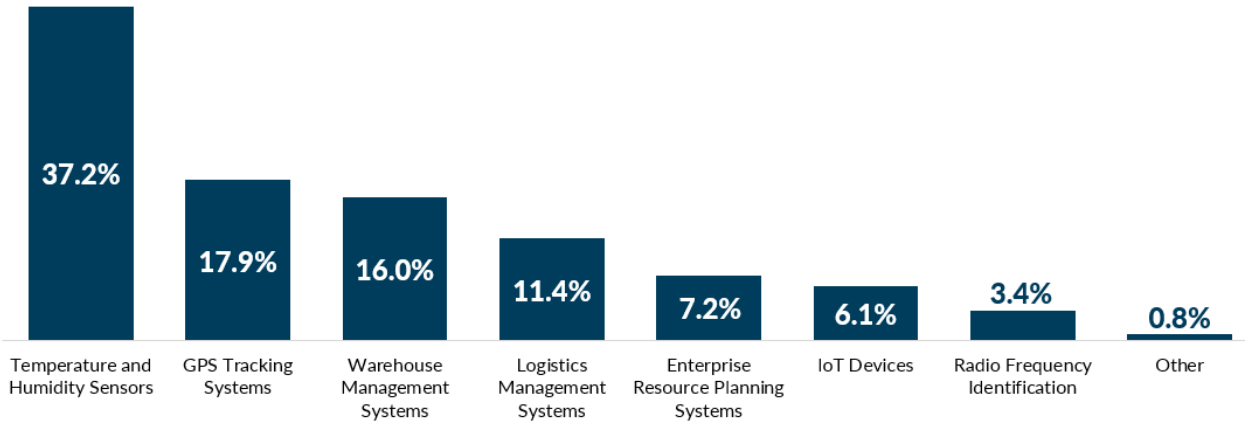
- All shipments are always monitored.
- Not applicable
- Not applicable in the context of shipping impact on product quality. If the question refers to thermal shipping qualification, the answer will be the first option
- We monitor every shipment

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.12 What systems does your company use to gather data? Please select all that apply.

Total Respondents: 107 *Total Responses: 263*



Data Gathering System	Responses	Percent
Temperature and Humidity Sensors	98	37.2%
GPS Tracking Systems	47	17.9%
Warehouse Management Systems	42	16.0%
Logistics Management Systems	30	11.4%
Enterprise Resource Planning Systems	19	7.2%
IoT Devices	16	6.1%
Radio Frequency Identification	9	3.4%
Other*	2	0.8%
Total	**263**	**100.0%**

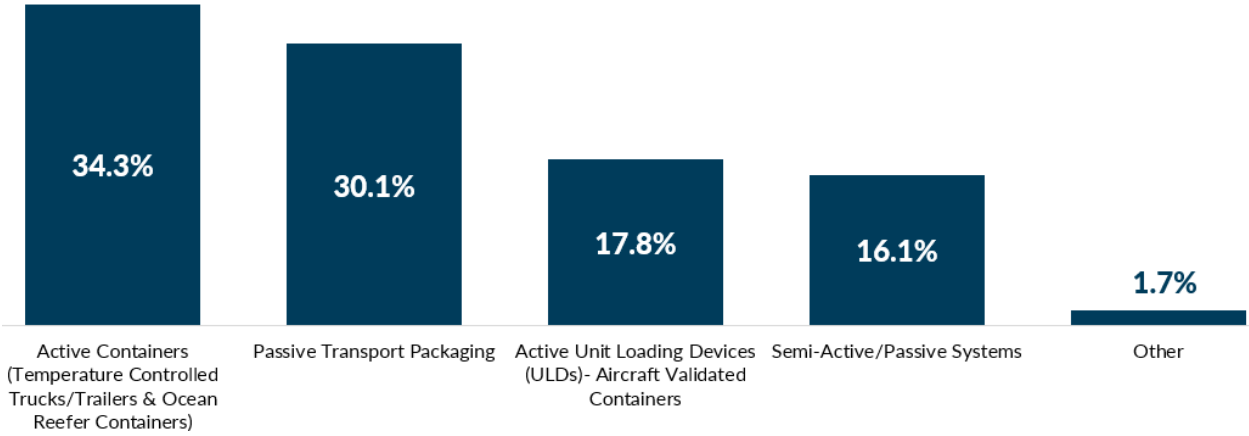
* Two responses were shared under the category "Other":

- Not applicable in the context of impact of shipping on product quality. The responses provided refer to thermal shipping qualification. Also use 4PL.
- Vibration tracking system

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.13 What type of shipping systems have you been using for real-world qualifications? Please select all that apply.

Total Respondents: 107 *Total Responses: 236*



Shipping System	Responses	Percent
Active Containers (Temperature Controlled Trucks/Trailers & Ocean Reefer Containers)	81	34.3%
Passive Transport Packaging	71	30.1%
Active Unit Loading Devices (ULDs)- Aircraft Validated Containers	42	17.8%
Semi-Active/Passive Systems	38	16.1%
Other*	4	1.7%
Total	**236**	**100.0%**

* Four responses were shared under the category "Other":

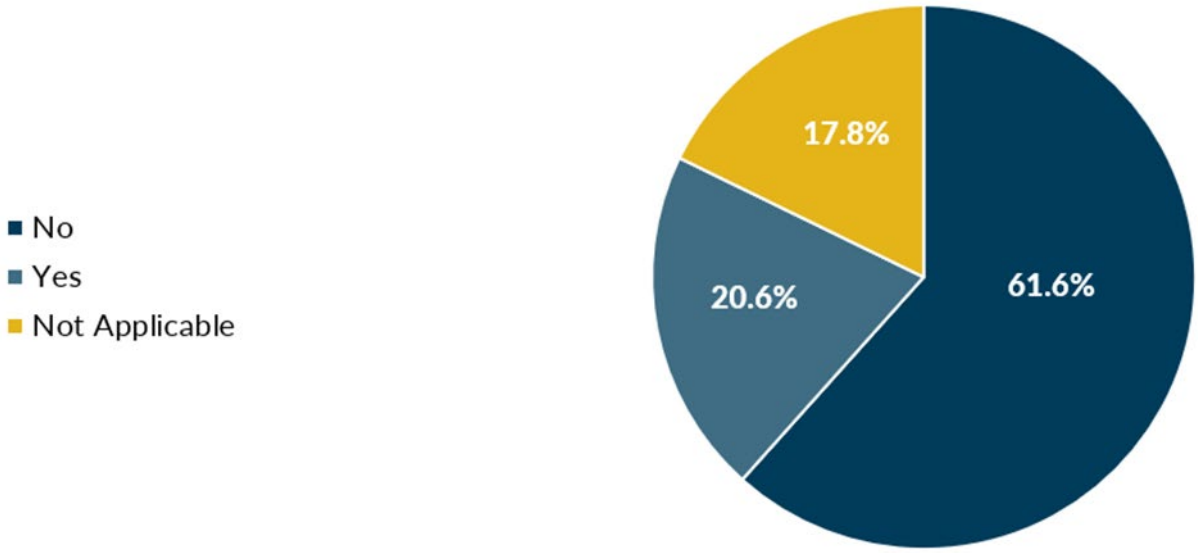
- N/A
- Not applicable in the context of impact of shipping on product quality. The responses provided refer to thermal shipping qualification.
- Reefer containers
- Thermo blankets

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.14 Has a regulatory agency challenged your company's placement of temperature recording devices within the shipping container for routine shipments?

Total Respondents: 107 *Total Responses: 107*

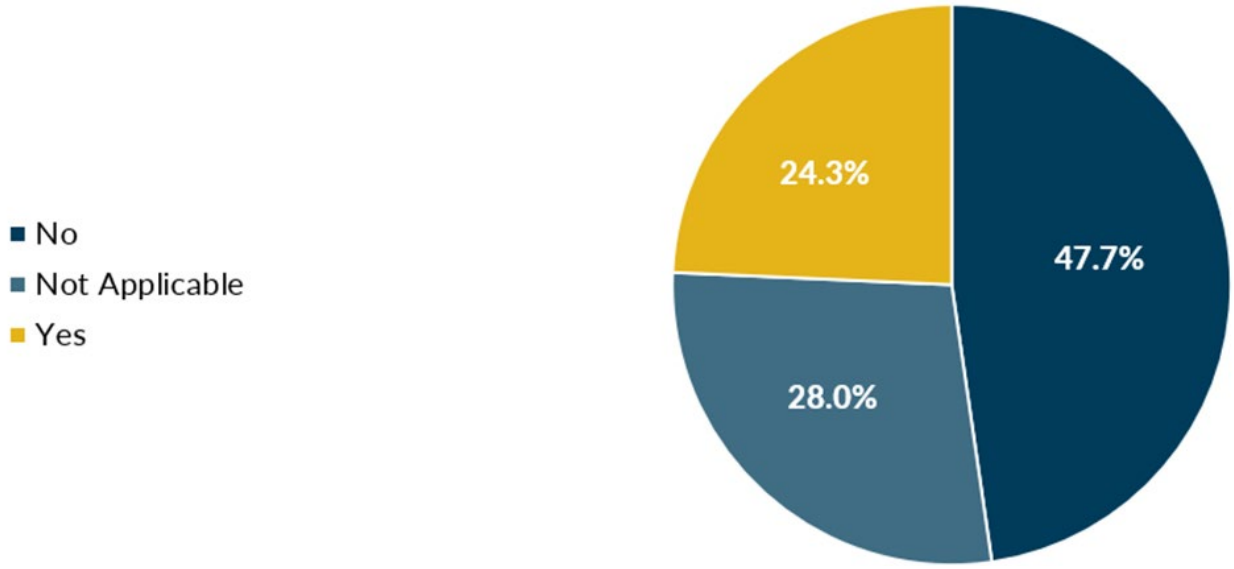


Regulatory Challenge	Responses	Percent
No	66	61.6%
Yes	22	20.6%
Not Applicable	19	17.8%
Total	**107**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.15 Has your organization received requests for real-world temperature data for shipments post-approval, prior to sending them to a specific region or country?

Total Respondents: 107 *Total Responses: 107*



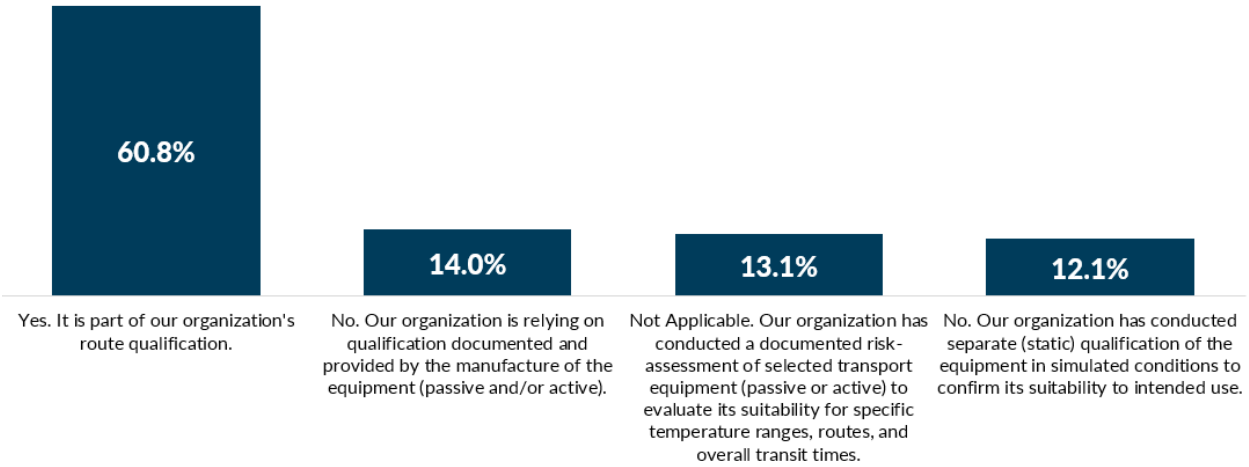
Real-World Temperature Data Requests	Responses	Percent
No	51	47.7%
Not Applicable	30	28.0%
Yes	26	24.3%
Total	**107**	**100.0%**

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.16 As part of route qualification, has your organization prepared and conducted qualifications for passive or active transport equipment?

Total Respondents: 107 *Total Responses: 107*

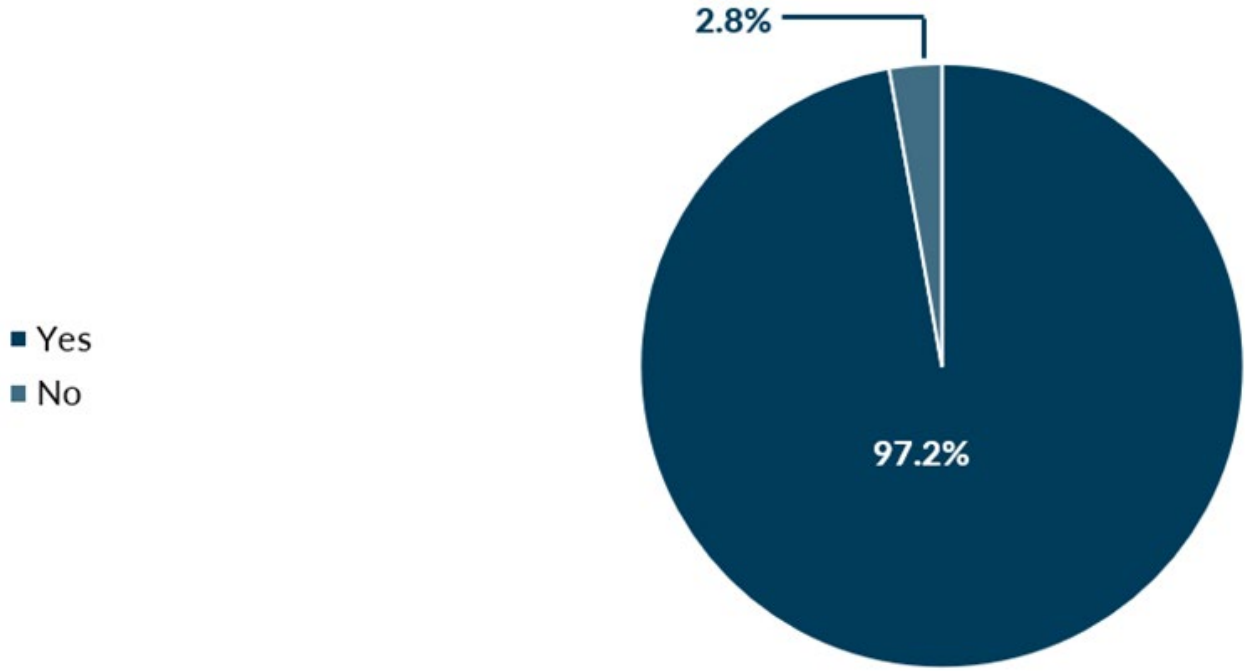


Prepared and Conducted Qualifications	Responses	Percent
Yes. It is part of our organization's route qualification.	65	60.8%
No. Our organization is relying on qualification documented and provided by the manufacture of the equipment (passive and/or active).	15	14.0%
Not Applicable. Our organization has conducted a documented risk-assessment of selected transport equipment (passive or active) to evaluate its suitability for specific temperature ranges, routes, and overall transit times.	14	13.1%
No. Our organization has conducted separate (static) qualification of the equipment in simulated conditions to confirm its suitability to intended use.	13	12.1%
Total	**107**	**100.0%**

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Q 2.17 Would it be beneficial to maintain our efforts in industry engagement, benchmarking, and collaboration with agencies concerning transport qualification studies?

Total Respondents: 107 *Total Responses: 107*



Maintain Efforts	Responses	Percent
Yes*	104	97.2%
No	3	2.8%
Total	**107**	**100.0%**

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

* One hundred and four respondents answered "Yes" that it would be beneficial to maintain our efforts in industry engagement, benchmarking, and collaboration with agencies concerning transport qualification studies. These respondents were asked, “Please share any comments and/or feedback you would like to provide to the authors of this survey.” Forty-nine respondents provided the following responses:

- Better alignment of practices and standardization of approaches would be helpful for industry.
- Better to share the information and take the feedback
- Clear guidelines need to set which have clarity to regulatory agencies and manufacturer.
- Company operates around the world. We use many approaches for different markets and regulations. It is hard to know of all regulatory feedback received as it is performed regionally.
- consider updating Technical Reports issued in the years past
- Foster Global Approach on TV requirements.
- Freeze-Thaw results should be considered on the product integrity. Kinetic temperature calculations and stability extrapolation should be required to estimate product effects.
- Good luck
- Good to understand current trends/expectations.
- Having an active engagement with agency will help to understand the expectations and allign the requirements well before
- Having multiple variables, if the survey specific to those would be more accurate.
- Help unify standards
- I believe its important to get alignment about regular practices in other companies and expected results from the regulatory agencies.
- I think this is timely and a good discussion in the pharma space. Transportation studies and qualifications are changing based on Covid and higher costs in biologics. We'd be happy at PSN Labs to support in these efforts to harmonize guidance.
- I would like to help PDA build Guidances and documents, templates, etc. To help the industry.
- I would like to see the results of this survey
- It would be good to have aligned expectations to follow
- It would be helpful to know that the guidance documents developed and provided are acceptable to the agencies.
- Keep updating the standards - there so many technologies to improve and minimize the transportation of goods.
- Laboratory shipping studies are conducted to de-risk real world shipments in the global supply chain.
- Missing (real-world/ practical) standards
- Note that the answers in this survey are focused on DP rather than intermediates we look forward to the summary of this benchmark
- One key aspect not covered by the survey is the potential economical loss for a temperature excursion. More specifically, how to resolve an exursion, for example, of 2 degrees C for 4 hours.
- PDA is go to non-government organizations which contributes high values to industry. Benchmarking would be beneficial to achieve consistent approach which provide ROI to industry.
- PDA should issue a Technical Report providing country and mode (air, ground, ocean) specific recommendations and examples for compliance.
- PDA technical reports provide indispensable information regarding good distribution practices
- Please consider ASTM D6653/D6653M − 13 as a reference to design a Cold chain / Shipping validation
- Please consider the transportation of microbiological samples

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

- Product stability that can be impacted by temperature excursions require validation throughout the supply chain process. Providing guidance to the industry is warranted.
- Provide correlation data between real world and simulation data to support the use of simulation data as representative of real world transport
- Regulators have pushed back on using laboratory studies for product quality when standard package test methods are used. There are regulator/industry expectations for how to simulate transportation conditions (real time, concurrent, realistic) that are not currently covered by any industry guidance.
- So much to say and not enough time to say it. Since there are no labs that I am aware of that can perform shock/vibration/drop testing under controlled (2 to 8C, -20C, -80C) environmental conditions, my preferred approach is to perform the shock/vibration/drop testing using product/packages that have been conditioned to the appropriate temperature, and tested immediately after removal from the conditioning. This allows the assessment of how temperature can influence package performance (things tend to break easier at cold temperatures). Those tested packages are then used to evaluate (essentially confirm) the thermal performance of the package. Cracks and other defects created from shock/vibration/drop that impact thermal performance are difficult to see with the naked eye (and pressure testing for 'leaks' is not possible). The standard methodologies employed in aforementioned testing have been thoroughly vetted by the industry and many in place for decades. The results from this 'simulated' testing are far more reliable than any real world 'confirmation' (kick it down the stairs or. ship it and see what happens) studies. A real world shipment is a snapshot and never fully representative. Also, most do not include GPS tracking and/or shock and vibration sensors to understand what the package encountered. I would not perform 'real world' confirmation studies if not expected by regulatory agencies. FYI - Regulators do not tend to inquire about package performance testing as they do not understand the topic. Never in 33 years (shock/vib or temperature) for PAI, planned or unplanned audits. This is true for Serialization as well.
- Survey results are always a useful guide when developing an approach to an area of new regulatory concern.
- Thank you for initiating this survey to determine what approaches are being used across the industry!
- Thank you for this initiative. We really appreciate it. When completing the survey, we noted that some questions would be answered differently depending on whether they refer to shipping impact on product quality or to thermal shipping qualification. This made the response to the survey challenging for some questions. For those questions that were unclear, we responded based on PQ. For those questions that had a "other" option, we provided the approach to responding to the question in that field. For questions 20-22 we responded based on thermal shipping qualification. Overall, maybe worth testing the survey before launching to identify potential ambiguities that may reduce the clarity of the responses and confuse the assessment. Thank you!!!
- The regulatory guidance for transport validation intended for BLA submission is not clear or consistent, leading to different interpretations as well as close collaboration with the regulatory agencies before submission. To start, the terms validation, qualification, and verification in transport area is not clear or well defined, leading to different interpretation (although PDA has released a well defined language definition on the terms). Qualification and validation are used exchangeable, whereas other guidance have committed to only use the term qualification assuming that a validation process is not possible in a such variable conditions during transport. Minimal requirements for cold chain before submission have not yet been established or well defined. Problems such as: Can qualification of devices be justified by manufacturers? Can a summer qualification be used as guidance for qualifying cold chain transport and be sufficient as worse-case? Can 3PL agencies and regulatory authorities have standards or common sense and assured preferences for minimal BLA requirements for passive and active TPSs to agree upon? Biologicals

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

focusing in cell based DPs are far behind in regulatory guidance. Therefore there is a must need for engagement in larger rounds of collaborations counting on industry, regulatory, and 3PL agencies. When considering African manufacturing sites or China drug product intermediates might be consider during transport. So far the transport guidelines available focus in APIs, Excipients or final DP. But how about Bulk/DP intermediate transport? No clear guidance in how much qualification/validation studies have to be covered for intermediates that for example can be integrated into aseptical processes, foreseen as very critical in transport risk assessments. How can AI systems improve and be implemented in the data collection and assure temperature monitoring in a faster manner? Can we design a guidance of available data/AI tools and courses to train personnel on it? Can agencies agree what is the minimal requirement for temperature monitoring data? Can cycling studies be reduced and reliability of modeling of temperature during temperature excursion can be integrated, even if MKTs is not the best model for the DP formulation? Transport validation sections still quite vague to define strategy in pharma companies and this leads to overcompensation in experimental database collection that could be reduced if clarifications and assurance by regulatory agencies become more clear with focus in countries specificities Eg: South America, Africa and Asian countries. Looking forward to an update guideline in this topic from basic to complexity concepts for the transportation chains.

- There is not a one place to go to/guidance which can be applied both to design verification and shipping validation. Terms are misunderstood and often misinterpreted. Hence discussing this topic is important.
- There should be set standards that any organization can utilize that are based on real-time validation and qualification activities.
- These efforts, if for the right reasons, are always beneficial to the industries and the patients being served. Based on my experience, the largest gap between real world and simulation is the actual physical handling of packaging systems throughout a shipments routing.
- This is a complex and misunderstood topic and industry would benefit from having more clarity on the topic.
- To release latest standards; Encourage test labs to participate in panel discussions; Share the success stories -Approvals, Positive feedback from Agencies
- Transport qualification should focus on confirming product robustness to transport hazards separate from thermal packaging qualification which should be focused on maintaining temperature control and physical integrity of associated packaging. Furthermore, confirmation of product robustness through global distribution through real-world shipping studies in adequate for many reasons, including the inability to control the most hazards in the field and ensure that worst-case exposure levels (i.e., the operating space) are properly tested. "Simulated" transport studies executed in a controlled-lab setting are well accepted by regulators globally, and ensure that the study can be designed/executed with worst-case hazard levels and well-controlled through execution. Concurrent hazard testing & non-accelerated vibration profiles are more realistic than sequential testing sequences with accelerated vibration profiles, and should be especially considered for biologics that are generally more sensitive than small molecule products. We've seen many companies over the years get regulatory pushback/rejection because of sequential testing sequences with accelerated vibration profiles like ASTM D4169.
- Very good initiative which is needed so that transportation validation or practices shall be uniformly drafted as a guidance to the industry, so that even small scale industries
- We are anxious to see the survey results from all respondents.
- We would appreciate if you could provide a report of the survey outcomes, including our company's position versus the market/industry.
- Whatever you like.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

- Would be good to have definitive guidance especially around use of simulated studies which seem undesired.
- Would be great to get a pulse on FDA expectations around transportation validation as it pertains to simulated vs real-world shipments, and expectation around Product Quality Testing post-shipment/simulations.
- Yes

 © 2025 Parenteral Drug Association, Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Terms of Usage for PDA Electronic Publications (Technical Reports, Books, etc.)

An **Authorized User** of this electronic publication in the purchase.

Authorized Users are permitted to do the following:

- Search and view the content of the PDA Electronic Publication
- Download the PDA Electronic Publication for the individual use of an Authorized User
- Print the PDA Electronic Publication for the individual use of an Authorized User
- Make a reasonable number of photocopies of a printed PDA Electronic Publication for the individual use of an Authorized User

Authorized Users are not permitted to do the following:

- Allow anyone other than an Authorized User to use or access the PDA Electronic Publication
- Display or otherwise make any information from the PDA Electronic Publication available to anyone other than an Authorized User
- Post the PDA Electronic Publication or portions of the PDA Electronic Publication on websites, either available on the Internet or an Intranet, or in any form of online publications without a license from PDA, Inc.
- Transmit electronically, via e-mail or any other file transfer protocols, any portion of the PDA Electronic Publication
- Create a searchable archive of any portion of the PDA Electronic Publication
- Use robots or intelligent agents to access, search and/or systematically download any portion of the PDA Electronic Publication
- Sell, re-sell, rent, lease, license, sublicense, assign or otherwise transfer the use of the PDA Electronic Publication or its content
- Use or copy the PDA Electronic Publication for document delivery, fee-for-service use, or bulk reproduction or distribution of materials in any form, or any substantially similar commercial purpose
- Alter, modify, repackage or adapt any portion of the PDA Electronic Publication
- Make any edits or derivative works with respect to any portion of the PDA Electronic Publication including any texts or graphics
- Delete or remove in any form or format, including on a printed article or photocopy, any copyright information or notice contained in the PDA Electronic Publication
- Combine any portion of the PDA Electronic Publication with any other material

To License or purchase Reprints

- Licensing: Site licenses and licenses to distribute PDA Electronic Publication can be obtained for a fee.
  To learn more about licensing options and rates, please contact:
  Whitney Alexis, Publications Manager, +1 (301) 656-5900.
  Written correspondence can be sent to: PDA, Inc. Attn: Whitney Alexis, 4350 East West Highway, Suite 150, Bethesda, MD 20814.
- Reprints: Reprints of PDA Electronic Publication can be purchased for a fee.
  To order reprints, please contact:
  Whitney Alexis, Publications Manager, +1 (301) 656-5900.
  Written correspondence can be sent to: PDA, Inc. Attn: Whitney Alexis, 4350 East West Highway, Suite 150, Bethesda, MD 20814.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

PDA Survey: Transportation Validation Benchmarking

About PDA Surveys

PDA conducts benchmarking surveys to assess the state of the industries we serve. We explore such topics as the current challenges in regulatory science, quality, analytical, and manufacturing site operations, and processes of pharmaceuticals and biotechnology products. These surveys are research tools used to gain information, usually from a targeted audience within PDA's areas of interest. The results, which are published or presented at meetings, represent the perspective of PDA survey respondents. Findings are used to focus PDA support activities or to contribute to the development of technical documents, courses, and meetings, fulfilling the professional needs and bringing additional value to PDA members.

While PDA goes to great lengths to ensure each Survey is of the highest quality, all readers are encouraged to contact PDA about any scientific, technical, or regulatory inaccuracies, discrepancies, or mistakes that might be found in any of the documents. Readers can email PDA at: TRfeedback@pda.org.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

PDA Survey: Transportation Validation Benchmarking

Sanctioning Advisory Board: SAB

Chair
Gabriele Gori
Chiesi Farmaceutici, S.p.A.

Vice-Chair
Ivy Louis
VIENNI Training & Consulting LLP

Frederic Ayers
ValSource, Inc.

Tiffany Baker
ValSource, Inc.

Gregory Bassett
Bristol Myers Squibb

Emily Cheah, PhD
Charles River Laboratories

Thomas Damratoski
CivicaRX

Cheryl Essex
Sanofi

David Exline
Gateway Analytical, LLC.

Rhonda Ezell
Alkermes

Mauro Giusti
Eli Lilly Italia

Rishikesh Jaiwant
Glaxo Smith Kline

Austin Kuo
Eli Lilly & Company

Beth Kirschenheiter
Bristol Myers Squibb

Gitte Letting
Novo Nordisk A/S

Bruce Loxley
Glaxo Smith Kline

Vincent O'Shaughnessy
Amgen

Ken Paddock
Merz Therapeutics

Julian Petersen
Groninger & Co. GMBH

Jim Polarine Jr.
STERIS Co.

Alexander Stoll, PhD
Fresenius-Kabi

Kristin Valente
Merck & Co., Inc.

Rick Watson
Merck & Co., Inc.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

PDA Survey: Transportation Validation Benchmarking

PDA Officers

Chair
Anil Sawant, PhD
Merck & Co., Inc.

Chair-Elect
Melissa Seymour, MBA
Bristol Myers Squibb

Secretary
Bettine Boltres, PhD
West Pharmaceutical Services

Treasurer
Emma Ramnarine, PhD
Boehringer Ingelheim

Immediate Past-Chair
Sue Schniepp
RCA

President
Glenn E. Wright
PDA

PDA Publication Production and Editing

Jessie Lindner
Project Manager, SAB

Josh Eaton
Senior Director, Scientific & Regulatory Affairs

Walter Morris
Senior Director, Publishing and Press Relations

Marilyn Foster
Technical Editor/Writer

PDA Directors

Marcia Baroni
Emergent BioSolutions, Inc.

Lisa Bennett
LB Consulting Australia Pty Ltd.

Cristiana Campa, PhD
Glaxo Smith Kline

Andrew Chang, PhD
Novo Nordisk, Inc.

Cylia Chen Ooi, MA
Amgen

Marc Glogovsky, MS
ValSource, Inc.

Andrew Hopkins
Lachman Consultants

Ivy Louis, MBA
VIENNI Training & Consulting LLP

Amy McDaniel, PhD
Bristol Myers Squibb

Morten Munk
FUJIFILM Diosynth Biotechnologies

Brigitte Reuter-Haerle
Vetter Pharma International GmbH

Osamu Shirokizawa
Life Scientia, Ltd.

Licensed to Kuo, Li-Hung/Amaran Biotechnology, Inc.: Copying and Distribution Prohibited.

Parenteral Drug Association®

Connecting People, Science and Regulation®

pda.org

PDA Global Headquarters
4350 East West Highway, Suite 600
Bethesda, MD 20814 USA
TEL +1 (301) 656-5900
FAX +1 (301) 986-0296

PDA Europe
Am Borsigturm 60
13507 Berlin, Germany

PDA Asia Pacific
20 Bendemeer Rd. #04-02 B S
Bendemeer Centre, Singapore
339914

PDA®
Parenteral Drug Association